

# FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026 — ANSWER KEY (Version 1)

Chemistry — Class IX

Syllabus: Ch 1 — Nature of Chemistry | Ch 2 — Matter | Ch 3 — Atomic Structure | Ch 4 — Periodic Table | Ch 5 — Chemical Bonding | Ch 6 — Stoichiometry | Ch 7 — Electrochemistry | Ch 8 — Energetics | Ch 9 — Chemical Equilibrium

**Total Marks: 65**

**Time Allowed: 2 Hours 30 Minutes**

**Version: 1**

## SECTION A — MCQ Answer Key (12 × 1 = 12 Marks)

| #  | Question (shortened)                        | Answer   | Explanation                                                                                                                                                                                                                                          |
|----|---------------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Branch dealing with carbon compounds        | <b>C</b> | <b>Organic chemistry</b> is the study of carbon compounds (hydrocarbons and their derivatives). Inorganic chemistry covers all other elements; physical chemistry studies structure and energy changes; analytical chemistry identifies composition. |
| 2  | Solid directly to gas                       | <b>A</b> | <b>Sublimation</b> is the direct solid-to-gas change (e.g., iodine, dry ice, naphthalene, ammonium chloride). Deposition is the reverse (gas to solid); fusion is solid to liquid; condensation is gas to liquid.                                    |
| 3  | Isotopes have                               | <b>B</b> | <b>Same protons, different neutrons.</b> Isotopes have the same atomic number but different mass numbers, e.g., $^{12}\text{C}$ , $^{13}\text{C}$ , $^{14}\text{C}$ .                                                                                |
| 4  | Max electrons in M shell ( $n=3$ )          | <b>D</b> | <b>18.</b> Shell capacity = $2n^2 = 2(3)^2 = 18$ . So K=2, L=8, M=18, N=32.                                                                                                                                                                          |
| 5  | Atomic radius across a period               | <b>A</b> | <b>Decreases.</b> Across a period the nuclear charge rises while electrons are added to the same shell, so the increased pull draws the shell inward.                                                                                                |
| 6  | Complete transfer of electrons              | <b>C</b> | <b>Ionic bond</b> (e.g., NaCl). Covalent = sharing of electrons; coordinate = both shared electrons from one atom; metallic = a sea of delocalized electrons.                                                                                        |
| 7  | Particles in one mole                       | <b>D</b> | <b><math>6.022 \times 10^{23}</math>,</b> Avogadro's number, the number of representative particles in one mole of any substance.                                                                                                                    |
| 8  | Empirical formula of $\text{C}_6\text{H}_6$ | <b>A</b> | <b>CH.</b> Dividing the subscripts by their highest common factor (6): $\text{C}_6\text{H}_6 \rightarrow \text{CH}$ , the simplest whole-number ratio.                                                                                               |
| 9  | Oxidation (electron terms)                  | <b>B</b> | <b>Loss of electrons.</b> Remember OIL RIG: Oxidation Is Loss, Reduction Is Gain (of electrons).                                                                                                                                                     |
| 10 | $\Delta H$ for an exothermic reaction       | <b>A</b> | <b>Negative.</b> Heat is released and the products lie at lower energy than the reactants, so $\Delta H < 0$ (e.g., combustion).                                                                                                                     |
| 11 | Forward vs reverse rate at equilibrium      | <b>B</b> | <b>Equal to.</b> At dynamic equilibrium the forward and reverse rates are equal, so the concentrations of reactants and products remain constant.                                                                                                    |
| 12 | Rusting of iron                             | <b>D</b> | <b>Corrosion.</b> In air and moisture, iron is oxidized to hydrated iron(III) oxide (rust). The corrosion of iron specifically is called rusting.                                                                                                    |

## SECTION B — Short Question Answers (11 × 3 = 33 Marks)

**2(i) Define chemistry; THREE branches** 3 marks

**Chemistry** is the branch of science that deals with the composition, structure, properties, and changes of matter.

1 mark

**Three branches:** 2 marks

- **Organic chemistry:** study of carbon compounds.
- **Inorganic chemistry:** study of all the elements and their compounds except most carbon compounds.
- **Physical chemistry:** study of the physical principles and energy changes that govern chemical reactions. (Biochemistry, analytical, and industrial chemistry are also acceptable.)

**OR****Science vs Technology vs Engineering** 3 marks

- **Science:** systematic knowledge of the natural world gained through observation and experiment, e.g., discovering that water is  $H_2O$ . 1 mark
- **Technology:** the application of scientific knowledge to make useful products and tools, e.g., a water-purification filter. 1 mark
- **Engineering:** the design and large-scale construction of structures, machines, and processes using science and technology, e.g., building a water-treatment plant. 1 mark

**2(ii) Element vs Compound vs Mixture** 3 marks

- **Element:** a pure substance made of only one kind of atom; cannot be broken down by chemical means. e.g., oxygen ( $O_2$ ), gold (Au). 1 mark
- **Compound:** a pure substance formed when two or more elements combine chemically in a fixed ratio; its properties differ from those of its elements. e.g., water ( $H_2O$ ). 1 mark
- **Mixture:** two or more substances physically combined in any ratio; components keep their own properties and can be separated physically. e.g., air, salt solution. 1 mark

**OR****Solution vs Suspension vs Colloid** 3 marks

- **Solution:** a homogeneous mixture, particle size  $< 1$  nm; does not scatter light or settle. e.g., salt in water. 1 mark
- **Suspension:** a heterogeneous mixture, particles  $> 1000$  nm, visible, settle on standing. e.g., muddy water, chalk in water. 1 mark
- **Colloid:** particle size 1-1000 nm; looks homogeneous but scatters light (Tyndall effect) and does not settle. e.g., milk, fog. 1 mark

**2(iii) Atomic number, mass number;  $^{35}_{17}Cl$**  3 marks

- **Atomic number (Z):** the number of protons in the nucleus of an atom. 0.5 mark
- **Mass number (A):** the total number of protons and neutrons in the nucleus. 0.5 mark

For  $^{35}_{17}Cl$ : protons =  $Z = 17$ ; electrons = **17** (neutral atom); neutrons =  $A - Z = 35 - 17 = 18$ . 2 marks

**OR****Isotopes of hydrogen** 3 marks

**Isotopes** are atoms of the same element with the same atomic number but different mass numbers (different numbers of neutrons). 1 mark

- **Protium ( $^1_1H$ ):** 1 proton, 0 neutrons. 0.5 mark
- **Deuterium ( $^2_1H$ ):** 1 proton, 1 neutron. 0.5 mark
- **Tritium ( $^3_1H$ ):** 1 proton, 2 neutrons (radioactive). 1 mark

**2(iv) Electronic configuration of Na and Cl** 3 marks

- **Sodium ( $Z=11$ ):** 2, 8, 1  $\rightarrow$  valence electrons = **1**. 1.5 marks
- **Chlorine ( $Z=17$ ):** 2, 8, 7  $\rightarrow$  valence electrons = **7**. 1.5 marks

**OR****Modern periodic law; groups and periods** 3 marks

- **Modern periodic law:** the physical and chemical properties of elements are a periodic function of their atomic numbers. 1 mark
- **Groups:** the 18 vertical columns; elements in a group have the same number of valence electrons and similar chemical properties. 1 mark
- **Periods:** the 7 horizontal rows; the period number equals the number of occupied shells, and properties change gradually across a period. 1 mark

**2(v) Ionization energy and its variation** 3 marks

**Ionization energy** is the minimum energy required to remove the most loosely bound (outermost) electron from an isolated gaseous atom. 1 mark

- **Down a group:** it **decreases** — atomic size increases, so the outer electron is farther from and less tightly held by the nucleus. 1 mark
- **Across a period:** it **increases** — nuclear charge rises and size falls, so electrons are held more tightly. 1 mark

**OR****Metals vs Non-metals (any THREE)** 3 marks

| Property     | Metals                                  | Non-metals                   |
|--------------|-----------------------------------------|------------------------------|
| Lustre       | Shiny, lustrous                         | Dull (mostly)                |
| Conductivity | Good conductors of heat and electricity | Poor conductors / insulators |
| Malleability | Malleable and ductile                   | Brittle (when solid)         |
| Electrons    | Lose electrons → form cations           | Gain electrons → form anions |

1 mark for each correct difference (any three).

**2(vi) Ionic bond vs Covalent bond** 3 marks

- **Ionic bond:** the electrostatic force of attraction between oppositely charged ions formed by the complete transfer of electrons from a metal to a non-metal. e.g., NaCl. 1.5 marks
- **Covalent bond:** the bond formed by the mutual sharing of one or more electron pairs between two non-metal atoms. e.g., H<sub>2</sub>, Cl<sub>2</sub>. 1.5 marks

**OR****Formation of MgCl<sub>2</sub>** 3 marks

- Magnesium (2, 8, 2) has 2 valence electrons; chlorine (2, 8, 7) has 7. 1 mark
- Mg loses its 2 electrons to become Mg<sup>2+</sup>; each Cl atom gains 1 electron to become Cl<sup>-</sup>, so 2 Cl atoms are needed. 1 mark
- The oppositely charged Mg<sup>2+</sup> and two Cl<sup>-</sup> ions attract electrostatically to form ionic MgCl<sub>2</sub>. 1 mark

**2(vii) Molecular mass of H<sub>2</sub>SO<sub>4</sub>** 3 marks

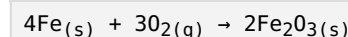
Molecular mass = 2(H) + 1(S) + 4(O)

$$= 2(1) + 32 + 4(16) = 2 + 32 + 64 = 98 \text{ amu}$$

1 mark setup, 1 mark substitution, 1 mark correct answer.

**OR****Moles in 36 g of water** 3 marks

- Molar mass of H<sub>2</sub>O = 2(1) + 16 = 18 g/mol. 1 mark
- moles = mass ÷ molar mass = 36 ÷ 18 = 2 mol. 2 marks

**2(viii) Balance Fe + O<sub>2</sub> → Fe<sub>2</sub>O<sub>3</sub>** 3 marks

Check: Fe → 4 = 4; O → 6 = 6. Balanced. 3 marks

**OR****Empirical vs Molecular formula** 3 marks

- **Empirical formula:** the simplest whole-number ratio of atoms of each element in a compound. e.g., CH (benzene), CH<sub>2</sub>O (glucose). 1.5 marks
- **Molecular formula:** the actual number of atoms of each element in one molecule. e.g., C<sub>6</sub>H<sub>6</sub> (benzene), C<sub>6</sub>H<sub>12</sub>O<sub>6</sub> (glucose). 1.5 marks

**2(ix) Oxidation and Reduction (electrons)** 3 marks

- **Oxidation:** loss of electrons by a species (its oxidation number increases). e.g.,  $\text{Na} \rightarrow \text{Na}^+ + \text{e}^-$ ;  $\text{Mg} \rightarrow \text{Mg}^{2+} + 2\text{e}^-$ . 1.5 marks
- **Reduction:** gain of electrons by a species (its oxidation number decreases). e.g.,  $\text{Cl}_2 + 2\text{e}^- \rightarrow 2\text{Cl}^-$ . 1.5 marks

**OR****Oxidation number of S in  $\text{H}_2\text{SO}_4$**  3 marks

- Let the oxidation number of S = x; H = +1, O = -2. 0.5 mark
- For a neutral molecule the sum is zero:  $2(+1) + x + 4(-2) = 0$ . 1 mark
- $2 + x - 8 = 0 \rightarrow x = +6$ . Sulphur has oxidation number +6. 1.5 marks

**2(x) Exothermic vs Endothermic reactions** 3 marks

- **Exothermic reaction:** releases heat to the surroundings;  $\Delta H$  is negative. e.g., combustion of methane, respiration, neutralization. 1.5 marks
- **Endothermic reaction:** absorbs heat from the surroundings;  $\Delta H$  is positive. e.g., photosynthesis, decomposition of  $\text{CaCO}_3$  on heating. 1.5 marks

**OR****Activation energy and catalysts** 3 marks

- **Activation energy ( $E_a$ ):** the minimum energy reactant molecules must possess to form the activated complex and convert into products. 1 mark
- A **catalyst** provides an alternative reaction path of **lower activation energy**. 1 mark
- This **increases the rate** of the reaction without itself being consumed, and it does not change  $\Delta H$ . 1 mark

**2(xi) Reversible reaction and dynamic equilibrium** 3 marks

- **Reversible reaction:** a reaction that proceeds in both forward and backward directions under the same conditions, shown by  $\rightleftharpoons$ . 1 mark
- **Dynamic equilibrium:** the state of a reversible reaction in a closed system where the forward rate equals the reverse rate, so concentrations remain constant. 1 mark
- **Example:**  $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$  (Haber process), or  $\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$ . 1 mark

**OR****Characteristics of dynamic equilibrium (any THREE)** 3 marks

- It is attained only in a **closed system**. 1 mark
- Forward and reverse reactions continue at **equal rates**; the reaction never stops (it is dynamic). 1 mark
- The **concentrations** of reactants and products stay constant, and macroscopic properties (colour, pressure) remain unchanged. 1 mark

**SECTION C — Long Question Answers ( $4 \times 5 = 20$  Marks)**

### 3(i) Bohr's atomic model; configuration of K and Ca 5 marks

#### Postulates of Bohr's model: 3 marks

- Electrons revolve around the nucleus only in certain fixed circular paths called **orbits / shells** (energy levels), each with a definite energy. 1 mark
- While an electron stays in a particular orbit it neither absorbs nor emits energy (these are **stationary states**). 1 mark
- An electron **absorbs energy** when it jumps to a higher orbit and **emits energy** (as light) when it falls to a lower orbit; the energy involved equals the difference between the two levels. 1 mark

#### Electronic configurations: 2 marks

- Potassium (Z=19):** 2, 8, 8, 1 → valence electrons = 1.
- Calcium (Z=20):** 2, 8, 8, 2 → valence electrons = 2.

OR

#### Subatomic particles; atomic number, mass number, isotope 5 marks

| Particle | Charge      | Relative mass            | Location                       |
|----------|-------------|--------------------------|--------------------------------|
| Proton   | +1          | 1 amu                    | Inside the nucleus             |
| Neutron  | 0 (neutral) | 1 amu                    | Inside the nucleus             |
| Electron | -1          | ~1/1836 amu (negligible) | Outside the nucleus, in shells |

3 marks for the comparison.

- Atomic number:** number of protons (e.g., carbon Z = 6).
- Mass number:** protons + neutrons (e.g., carbon-12, A = 12).
- Isotope:** atoms of the same element with different mass numbers (e.g.,  $^{12}\text{C}$  and  $^{14}\text{C}$ ). 2 marks

### 3(ii) Periodic trends across a period; electropositive vs electronegative 5 marks

#### Trends across a period (left → right): 3 marks

- Atomic size:** decreases — nuclear charge increases while electrons enter the same shell, increasing the inward pull. 1 mark
- Ionization energy:** increases — smaller size and greater nuclear charge hold the outer electrons more tightly. 1 mark
- Electronegativity:** increases — the atom attracts the shared bonding pair more strongly as charge rises and size falls. 1 mark

#### Electropositive vs electronegative: 2 marks

- Electropositive elements** lose electrons easily and form positive ions — metals such as Na, K.
- Electronegative elements** gain electrons and form negative ions — non-metals such as Cl, O.

OR

#### Ionic bond in NaCl; covalent bond in Cl<sub>2</sub>; differences 5 marks

- NaCl (ionic):** Na (2,8,1) loses 1 electron → Na<sup>+</sup>; Cl (2,8,7) gains that electron → Cl<sup>-</sup>; the ions attract electrostatically. 1.5 marks
- Cl<sub>2</sub> (covalent):** each Cl has 7 valence electrons; the two atoms **share one pair** of electrons so both complete their octet, forming a single covalent bond (Cl-Cl). 1.5 marks

| Property                | Ionic compounds                  | Covalent compounds                          |
|-------------------------|----------------------------------|---------------------------------------------|
| Physical state          | Usually solids                   | Often gases, liquids, or low-melting solids |
| Melting / boiling point | High                             | Low                                         |
| Electrical conductivity | Conduct when molten or dissolved | Generally non-conductors                    |
| Solubility              | Usually soluble in water         | Usually soluble in organic solvents         |

2 marks for any four differences.

### 3(iii) Mole and Avogadro's number; calculations 5 marks

#### Definitions: 2 marks

- **Mole:** the amount of a substance that contains as many particles as there are atoms in exactly 12 g of carbon-12, i.e.,  $6.022 \times 10^{23}$  particles. 1 mark
- **Avogadro's number:** the number of particles in one mole =  $6.022 \times 10^{23}$ . 1 mark

#### Calculations: 3 marks (1 each)

- (i) moles of  $\text{H}_2\text{SO}_4 = 49 / 98 = 0.5 \text{ mol}$  (molar mass = 98 g/mol)
- (ii) molecules in 0.5 mol  $\text{CO}_2 = 0.5 \times 6.022 \times 10^{23} = 3.011 \times 10^{23}$
- (iii) mass of 2 mol  $\text{NaOH} = 2 \times 40 = 80 \text{ g}$  (molar mass = 40 g/mol)

OR

### Balancing equations; molecular and formula mass 5 marks

#### Balanced equations: 3 marks (1 each)

- (i)  $\text{C}_3\text{H}_8 + 5\text{O}_2 \rightarrow 3\text{CO}_2 + 4\text{H}_2\text{O}$
- (ii)  $2\text{Al} + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2$
- (iii)  $2\text{KClO}_3 \rightarrow 2\text{KCl} + 3\text{O}_2$

#### Masses: 2 marks

- Glucose  $\text{C}_6\text{H}_{12}\text{O}_6 = 6(12) + 12(1) + 6(16) = 72 + 12 + 96 = 180 \text{ amu}$ . 1 mark
- Calcium carbonate  $\text{CaCO}_3 = 40 + 12 + 3(16) = 40 + 12 + 48 = 100 \text{ amu}$ . 1 mark

### 3(iv) Redox in $\text{Zn} + \text{CuSO}_4$ ; corrosion 5 marks

#### Oxidation, reduction, and agents: 3 marks

- **Oxidation** = loss of electrons; **reduction** = gain of electrons (both occur together in a redox reaction). 1 mark
- $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$  (loses electrons, is oxidized), so **Zn is the reducing agent**. 1 mark
- $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$  (gains electrons, is reduced), so  **$\text{CuSO}_4$  ( $\text{Cu}^{2+}$ ) is the oxidizing agent**. 1 mark

#### Corrosion: 2 marks

- **Corrosion** is the gradual eating away (oxidation) of a metal surface by reaction with air, moisture, or chemicals; for iron it is called rusting. 1 mark
- **Prevention (any two):** painting / greasing, galvanizing (zinc coating), electroplating, alloying (stainless steel), or cathodic protection with a sacrificial metal. 1 mark

OR

### Exothermic and endothermic reactions; dynamic equilibrium 5 marks

#### Exothermic vs endothermic: 3 marks

- **Exothermic:** releases heat,  $\Delta H$  negative; products lie at lower energy than reactants. Energy diagram falls from reactants to products. Example:  $\text{C} + \text{O}_2 \rightarrow \text{CO}_2 + \text{heat}$ . 1.5 marks
- **Endothermic:** absorbs heat,  $\Delta H$  positive; products lie at higher energy than reactants. Energy diagram rises from reactants to products. Example:  $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$  (on heating). 1.5 marks

Exothermic: Reactants ----\\_\_\_\_ (energy released,  $\Delta H < 0$ )  
Products

Endothermic: Products (energy absorbed,  $\Delta H > 0$ )  
Reactants \_\_/----

#### Dynamic equilibrium: 2 marks

- **Dynamic equilibrium:** the state at which forward and reverse reaction rates are equal and concentrations remain constant. 1 mark
- **Conditions (any two):** the system must be closed; temperature and pressure must be kept constant; the reaction must be reversible. 1 mark

— End of Answer Key —